the incumbent, according to our GP belief. Figure 4.12 shows this score across our search space together with our GP.

We observe some interesting behavior in the PoI scores:

- The region to the left of the incumbent (from 0 to 1) has relatively high PoI scores. This corresponds to the high mean predictions in this area.
- The region on the left of the plot has slightly lower scores. This is because the mean predictions are not as high, but there is enough uncertainty in this region that there's still considerable probability of improving.
- The region around 2 has PoI close to 0. As we saw, the predictive normal distributions of the points here mostly lie below the incumbent threshold.

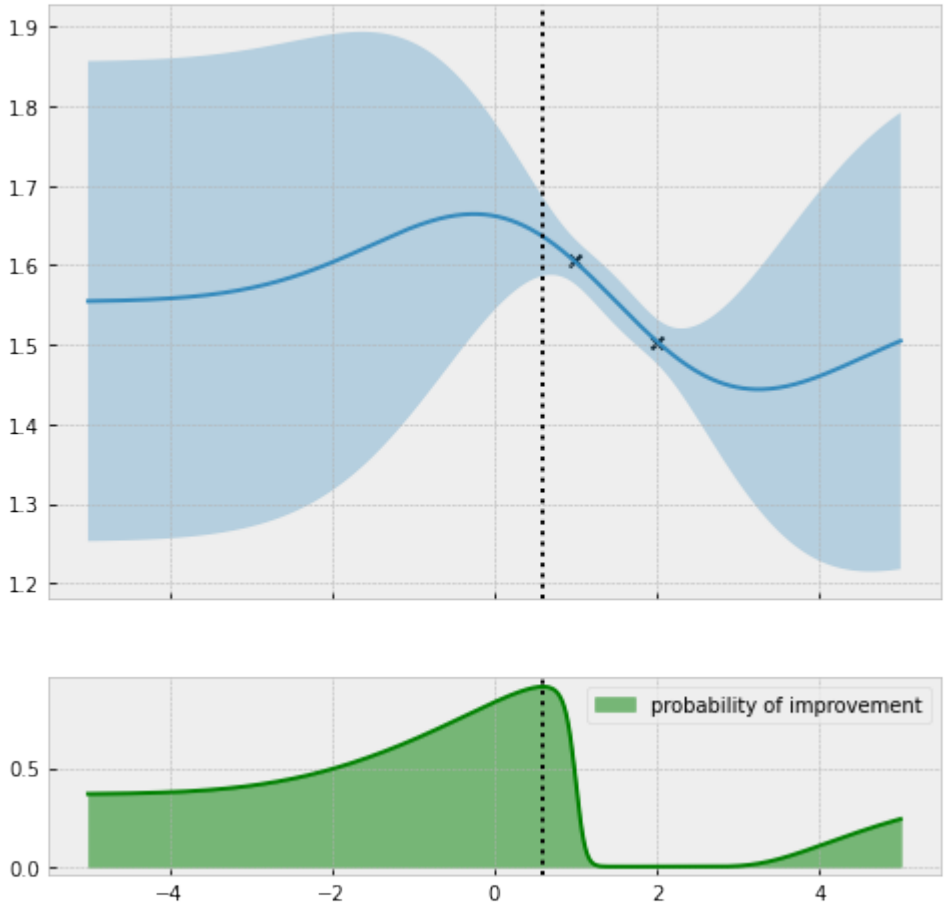


Figure 4.12 Gaussian process predictions (top) and Probability of Improvement (PoI) (bottom), where the dotted line indicates the point that maximizes the PoI score. This point is where we query the objective function at the next iteration of optimization.

Now, all that's left for us to do is to identify the point between -5 and 5 that has the highest PoI score, that is, the point that maximizes the probability of improving from the incumbent. As mentioned, we will take advantage of BoTorch's helper function `botorch.optim.optimize.optimize_acqf()`, which finds the point that maximizes the score of any BayesOpt policy. We do this using the following code, which is a part of the code that implements the BayesOpt loop:

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